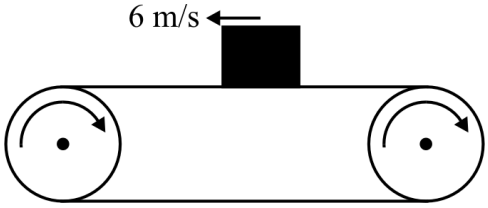
PART 1 : PHYSICS

SECTION-I (i)

1) A block lying on a long horizontal conveyor belt moving at a constant velocity receives a velocity 6 m/s relative to the ground in the direction opposite to the direction of motion of the conveyor. After t = 2 sec, the velocity of the block becomes equal to the velocity of the belt. The coefficient of friction between the block and the belt is 0.4. Then the speed of the conveyor belt is :-



(A) 1 m/s

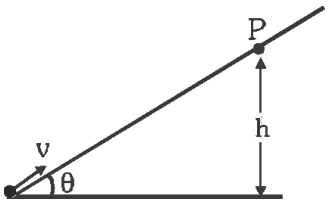
(B) 2 m/s

(C) 3 m/s

(D) 6 m/s

2)

A particle is thrown up with initial velocity v along a smooth fixed wedge as shown in figure. The particle reaches upto P and returns. The height (h) of point P is



(A) $\frac{v^2}{2g}$

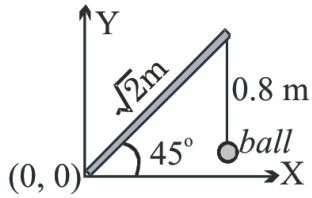
(B) $\frac{v^2}{2g\sin\theta}$

(C) $\frac{v^2}{2g}\sin^2\theta$

(D) $\frac{v^2}{2g}\cos\theta$

3) In the figure shown, a ball is suspended with a 0.8 m long cord from a rod of length √2 m. Find

the position vector of the ball w.r.t. origin.



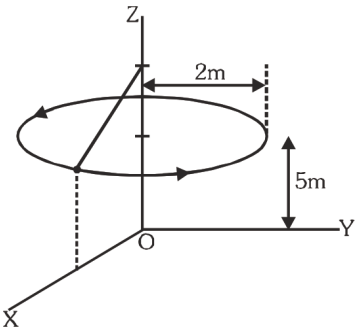
(A) $\hat{i} - 0.2\hat{j}$

(B) $\hat{i} + 0.2\hat{j}$

(C) $0.2\hat{i} - \hat{j}$

(D) $0.2\hat{i} + \hat{j}$

4) The upper end of the string of a simple pendulum is fixed to a vertical z-axis, and set in motion such that the bob moves along a horizontal circular path of radius 2 m, parallel to the xy plane, 5 m above the origin as shown. The bob has a speed of 3m/s. The string breaks when the bob is vertically above the x-axis, and it lands on the xy plane at a point (x, y), then-



(A) x = 2m

(B) x > 2m

(C) y = 2m

(D) y = 5m

SECTION-I (ii)

1) A beam of light parallel to main optical axis falls on a converging lens of focal length f_1 = 20 cm. Behind the lens, coaxially at some distance L, is diverging lens of focal length f_2 = – 20 cm. After passing through the diverging lens, the light focuses at a point 'A' 5 cm behind the lens. The position of lens is then interchanged :-

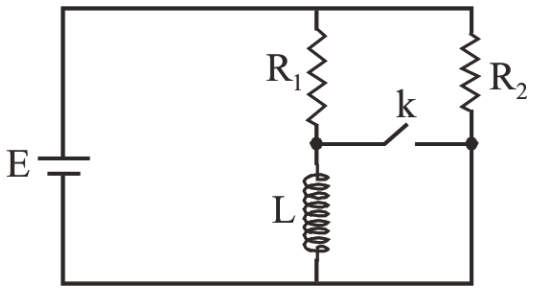
(A) The value of L is 16 cm.

(B) The value of L is 80/3 cm.

(C) Point A shifts by 40 cm after the lenses are interchanged

(D) Point A shifts by 125/3 cm after the lenses are interchanged

2) In the shown circuit, the resistors R_1 and R_2 have unequal (non-zero) resistance and L is an ideal inductor. The cell is ideal and all conducting wires have zero resistance. Initially the key k is open and the circuit is in steady state. Then after the key k is closed which of the following statement is



correct ?

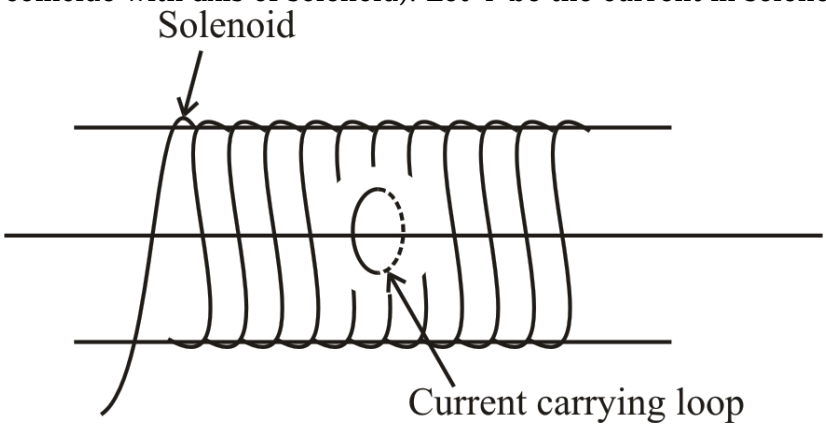
(A) Current through R_1 will not change

(B) Current through R_2 will not change

(C) Current through L will not change

(D) Current will pass through closed key

3) A single circular loop of wire with radius 0.02 m carries a current of 8.0 A. It is placed at the centre of a long solenoid that has length 0.65m, radius 0.080 m and 1300 turns. (Axis of circular loop coincide with axis of solenoid). Let 'i' be the current in solenoid.



(A) If i = 100 mA, the magnetic field at the center of coil can be zero.

(B) If i = 44 mA, the magnetic field at the center of coil can be zero.

(C) If i = 100 mA, the magnetic field at the center of coil can be $8\pi \times 10^{-5}$ T.

(D) If i = 100 mA, the magnetic field at the center of coil can be $16\pi \times 10^{-5}$ T.

SECTION-II

1)

A cyclist speeding at 2 m/s on a level road takes a sharp circular turn of radius 2m without reducing the speed. If minimum coefficient of static friction between tyres and road so that cyclist does not slip while taking the turn is $\frac{1}{n}$, find value of n. (Take g = 10 m/s^2)

2) If rain drops are falling with speed 12 m/s at angle of 30° with vertically for the observer at rest. Then if observer starts moving with constant acceleration 4 m/s^2 along horizontal. After time T observer observes rain drop again at an angle of 30° with vertical. Then find T.

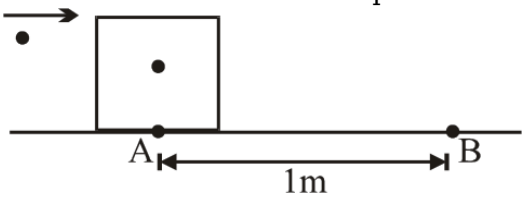
3)

A block of mass m is placed on a surface with a vertical cross section given by $y = \frac{x^3}{6}$. If the coefficient of friction is 0.5, the maximum height above the ground at which block can be placed without slipping is $\frac{x}{y}$. Find the value of (x + y).

4) Two vectors $\vec{A} = 2\hat{i} + 3\hat{j} - 4\hat{k}$ and $\vec{B} = 4\hat{i} + 8\hat{j} + x\hat{k}$ are such that the components of $\vec{B}$ along $\vec{A}$ is zero. Then the value of x is :

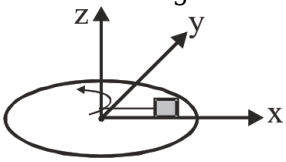
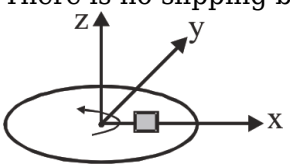
5) The radius R, of the fireball produced by an atomic bomb explosion may be assumed to depend on the density 'ρ' of the atmosphere, the time 't' elapsed since the explosion and the energy 'E' released by the bomb. Deduce the formula for R by dimensional method. If your answer is $R = kE^{\frac{1}{x}}t^{\frac{2}{y}}\rho^{-\frac{1}{z}}$ where k is dimensionless constant, fill the value of (x + y + z).

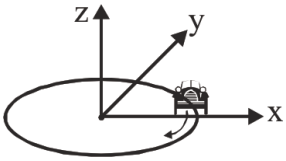
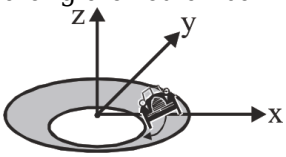
6) A bullet of mass 0.01 kg strikes a block of mass 0.09 kg and gets embedded in it. The block then slide along the surface and comes to rest at point B. The coefficient of friction varies as μ(x)=0.1x along the surface where AB = 1 m and x is measured from point A. Find the velocity (in m/s) with which the bullet hits the block.



SECTION-I (iii)

1) Mark the matching options for situations shown in respective figures of list-I at the instant when particle / car is located on the x-axis as shown :

	List-I		**List-II**
(P)	Block attached to string is moving along a circle on rough surface 	(1)	Force due to friction may have non-zero x-component
(Q)	Block is placed on a disc rotating with non-uniform angular velocity as shown below. There is no slipping between block & disc. 	(2)	Force due to friction may have non-zero y-component.

(R)	Car moving on ground along a circular horizontal track at a constant speed 	(3)	Force due to friction may have non-zero z-component.
(S)	Car moving on ground at constant speed along a circular banked track. 	(4)	Force due to friction may be zero.
		(5)	Centripetal force is only due to friction.

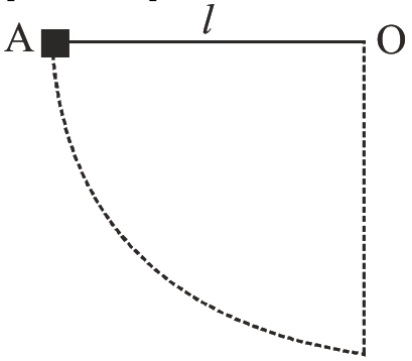
(A) P → 1,2,5;Q → 1,5;R → 1,3,4;S → 2

(B) P → 2;Q → 1,2,5;R → 1,5;S → 1,3,4

(C) P → 1,5;Q → 1,3,4;R → 1,2,5;S → 1,2

(D) P → 1,3,4;Q → 1,2,5;R → 1,5;S → 3,4

2) A block of mass m is tied with an inextensible light string of length □. One end of the string is fixed at point O. Block is released (from rest) at A. Find acceleration of particle during its motion in vertical plane at positions specified in list-I and match them with list-II. Given that A & O are at same horizontal level.



List-I		List-II	
(I)	Highest point	(P)	Acceleration is horizontal.
(II)	At lowest point	(Q)	Acceleration is vertically upwards.
(III)	at $\theta = \tan^{-1}(\sqrt{3})$ with vertical	(R)	Acceleration is vertically downwards.
(IV)	at $\theta = \tan^{-1}\frac{1}{\sqrt{3}}$ with vertical	(S)	Acceleration has both horizontal and vertical components.
		(T)	None of these.

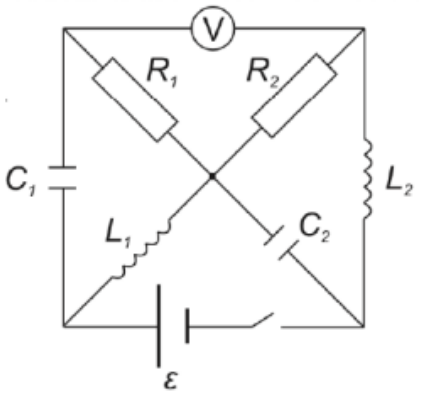
(A) I → P;II → R;III → Q;IV → S

(B) I → R;II → Q;III → S;IV → S

(C) I → R;II → Q;III → P;IV → S

(D) I → P;II → S;III → Q;IV → R

3) For the circuit shown in figure, $R_1 = 30\left(\sqrt{5}\right)$ ohm, $R_2 = 10\left(\sqrt{5}\right)$ ohm, $C_1 = C_2 = 18$ micro farad, and $L_1 = L_2 = 1$ mili henry. The electromotive force of the battery is E = 10 volt. The voltmeter connected in circuit is an ideal voltmeter. Initially the switch is closed and the system is



under steady state. Key opened at t = 0.

Column I		Column II	
P	Reading of voltmeter at steady state (just before t=0) is V_1 volt then $\|V_1\|$	1	10
Q	Reading of voltmeter just after key opened (just after t=0) is V_2 volt then $\|V_2\|$	2	100
R	Total heat dissipated in resistor R_1 after key opening.(after t=0) is H_1 micro joule. then H_1 is	3	1000
S	Total heat dissipated in resistor R_2 after key opening.(after t=0) is H_2 micro joule. then H_2 is	4	50
		5	20

(A) P → 1;Q → 2;R → 3;S → 4

(B) P → 2;Q → 5;R → 1;S → 4

(C) P → 1;Q → 5;R → 2;S → 3

(D) P → 1;Q → 2;R → 5;S → 3

PART 2 : CHEMISTRY

SECTION-I (i)

1) Cl–O bond order in perchlorate ion will be :-

(A) 1.33

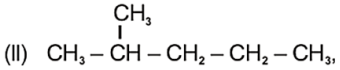
(B) 1.50

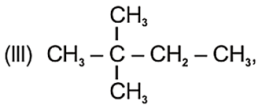
(C) 1.66

(D) 1.75

2) In the following compounds the decreasing order of boiling point is:

(I) $CH_3 - CH_2 - CH_2 - CH_2 - CH_2 - CH_3$,

(II) 

(III) 

(A) I > II > III

(B) I > III > II

(C) II > III > I

(D) III > II > I

3) Find the number of stereoisomers of 1,2-dihydroxy cyclopentane :-

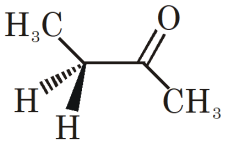
(A) 1

(B) 2

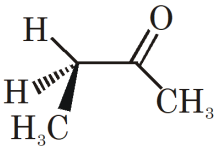
(C) 3

(D) 4

4) The correct statements about conformations X and Y of 2-butanone are :



(X)



(Y)

(a) **X** is more stable than **Y**

(b) **Y** is more stable than **X**

(c) Methyl groups in **X** are anti

(d) Methyl groups in **Y** are gauche

(A) a and d

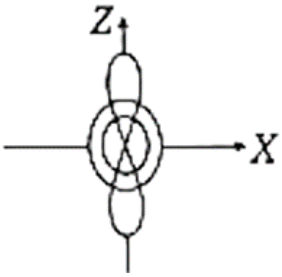
(B) a and c

(C) b and c

(D) a, c and d

SECTION-I (ii)

1) Consider the following atomic orbitals :



Which of the following statement(s) is/are correct regarding given orbital ?

(A) It has two nodal cones

(B) It has zero nodal plane

(C) Circular electron density is present in ' XY' plane

(D) It has different shape from other d-orbitals.

2)

The wave function for 2s orbital is given as $\psi = \left(\frac{1}{4\sqrt{2}}\right)\left(\frac{1}{a_0}\right)^{3/2}\left(2-\frac{r}{a_0}\right).e^{-r/2a_0}$

Where a_o = First Bohr's radius in H atom = 0.529 A°.

Read the given statement and pick out the correct statement(s).

(A) The number of radial nodes is equal to three

(B) The probability density is independent of direction

(C) The probability density of finding electron at nucleus is non-zero.

(D) The radial node occurs at a distance $2a_0$ from nucleus.

3) Select the correct statements:

(A) The molecularity of an elementary reaction indicates how many reactant molecules take part in the step

(B) The rate law of an elementary reaction can be predicted by simply seeing the stoichiometry of reaction

(C) The slowest elementary step in sequence of the reactions governs the overall rate of formation of product

(D) A rate law is often derived from a proposed mechanism by imposing the steady state approximation or assuming that there is a pre-equilibrium

SECTION-II

1) How many compounds are more reactive than ethene towards electrophilic addition reaction?

(I) $CH_3 - CH = CH_2$

(II) $H-\overset{\overset{O}{\|}}{C}-CH=CH_2$

(III) Cl – CH = CH_2

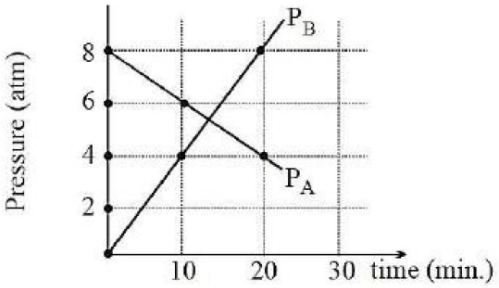
(IV) CH_3 – O – CH = CH_2

(V) H – C ≡ C – H

(VI) Ph – CH = CH_2

(VII) CH_2 = CH – CH = CH_2

2) In a chemical reaction where A(g) is converting to B(g), the following observation is made. What is

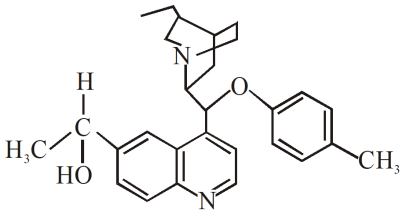


the time of completion for the reaction (in min)?

3) If x and y are total number of metal electrons which are present in non-axial and axial set of d-orbitals respectively in Ni cation of $[Ni(DMG)_2]$, then calculate value of $\frac{x^2+y^2}{x^2-y^2}$.

4) Consider complex salt $[Fe(H_2O)_6]_3[Fe(CN)_6]_2$
Find (x + y + z)
x = average number of unpaired electrons on each iron ion.
y = total number of 3d-orbitals involved in hybridisation for the formation of one molecule of above complex
z = average oxidation number of iron

5) The number of chiral carbons present in the molecule given below is _____ .



6) An organic compound P exists in two enantiomeric forms, which have specific optical rotation values $[\alpha] = \pm 100^o$. The optical rotation of a mixture of these two enantiomers is -50°. Calculate the percentage of that enantiomer which is lower concentration in the mixture.

SECTION-I (iii)

1) Matching the following :

	List-I (complex ion)		List-II (CFSE, hybridisation and magnetic nature)
(P)	$[Cr(NH_3)_6]^{3+}$	(1)	0.0 Δ_0, sp^3d^2-hybridisation
(Q)	$[Cu(NH_3)_6]^{2+}$	(2)	–0.6 Δ_0, sp^3d^2-hybridisation
(R)	$[Fe(H_2O)_6]^{3+}$	(3)	–1.2 Δ_0, d^2sp^3-hybridisation
(S)	$[IrF_6]^{3-}$	(4)	Diamagnetic
		(5)	Paramagnetic

Correctly matched among the following is :

(A) P → 3,5;Q → 2,5;R → 1,5;S → 4

(B) P → 4,5;Q → 2,3;R → 2,3;S → 4

(C) P → 3,5;Q → 2,5;R → 3,4;S → 3

(D) P → 2,5;Q → 3,5;R → 3;S → 4

2) Match the half-life of the reaction in List-I with the integrated rate equation in List-II (a = initial concentration of reactant and a – x – concentration of reactant at time 't'.)

List - I		List - II	
(P)	$t_{1/2} = \frac{0.586}{k}\sqrt{a}$	(1)	$k = \frac{x}{t}$
(Q)	$t_{1/2} = \frac{0.693}{k}$	(2)	$k = \frac{1}{t}\left[\frac{1}{a-x} - \frac{1}{a}\right]$
(R)	$t_{1/2} = \frac{a}{2k}$	(3)	$k = \frac{2}{t}\left[\sqrt{a} - \sqrt{a-x}\right]$
(S)	$t_{1/2} = \frac{1}{ka}$	(4)	$k = \frac{2.303}{t}\log\frac{a}{a-x}$
		(5)	$k = \frac{2}{t}\left[\frac{1}{\sqrt{a-x}} - \frac{1}{\sqrt{a}}\right]$

(A) P → 5;Q → 4;R → 1;S → 2

(B) P → 3;Q → 4;R → 1;S → 2

(C) P → 2;Q → 3;R → 4;S → 2

(D) P → 3;Q → 2;R → 4;S → 5

3) For the given galvanic cell at 25°C, Pt, H_2(g, 1 atm) | H^+(aq. 0.1M) || Cl^-(aq. 0.01 M) | Cl_2(g, 1 atm), match the list-I (change in concentration or quantity) with list-II (change in EMF).

(Given : $\frac{2.303RT}{F}$ = 0.06V, log2 = 0.3)

List-I		List-II	
(P)	pH of solution in anodic compartment change to pH = 5	(1)	– 0.03 V
(Q)	Concentration of ions in cathodic compartment increased by factor 10	(2)	0.03 V
(R)	Pressure of gas flown in anodic compartment increased by factor 10	(3)	– 0.06 V
(S)	Pressure of gas flown in cathodic compartment decreased by factor 10	(4)	0.24 V
		(5)	

(A) P → 3;Q → 4;R → 1;S → 2

(B) P → 4;Q → 3;R → 2;S → 1

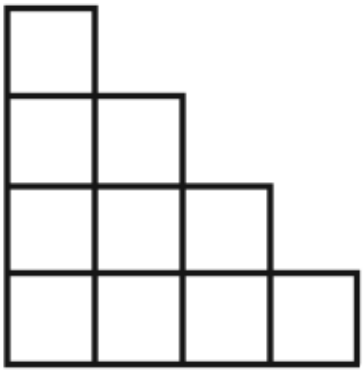
(C) P → 3;Q → 4;R → 2;S → 1

(D) P → 4;Q → 3;R → 1;S → 2

PART 3 : MATHEMATICS

SECTION-I (i)

1) The number of ways in which all the letters of the word 'BIGBOSS' can be placed in the squares of the figure shown, so that no row remains empty is equal to



(A) 95760

(B) 94640

(C) 97840

(D) 82120

2) The number of numbers, strictly between 5000 and 10000 can be formed using the digits 1,3,5,7,9 without repetition, is

(A) 6

(B) 12

(C) 120

(D) 72

3) If $\begin{vmatrix} x+1 & x+2 & x+a \\ x+2 & x+3 & x+b \\ x+3 & x+4 & x+c \end{vmatrix} = 0$ then line ax + by + c = 0 always pass through a fixed point P then P is

(A) (1, 1)

(B) (1, 2)

(C) (1, –2)

(D) (2, 1)

4) Let $A=\begin{bmatrix}\sqrt{3} & 1 & 0\\ 1 & -\sqrt{3} & 0\\ 0 & 0 & 2\end{bmatrix}$ and d = det($2A^T + AA^T$ + adjA) then $\sqrt{d}$ is

(A) 1

(B) 2

(C) 8

(D) 16

SECTION-I (ii)

1) Let f(x) = $\cos^2 x\,(e^{\tan x})$ $x \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$ then

(A) f ′(x) has a point of local minima at $x = \frac{\pi}{4}$

(B) f ′(x) has a point of local maxima in interval $\left(-\frac{\pi}{4}, 0\right)$

(C) f ′(x) has exactly two point of extrema in $\left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$

(D) f ″(x) has no root in $\left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$

2) Let b ∈ R–{0}, suppose f : R → R^+ is a differentiable function such that f(0) = 1. If the derivative f' of f satisfies the equation $f'(x) = \frac{f(x)}{\sqrt{b^2 - x^2}}$ for all x ∈ R, then which of the following statements is/are true ?

(A) Monotonic behaviour of f(x) depends on b.

(B) Monotonic behaviour of f(x) is independent of b.

(C) f(x) . f(–x) = 1 for x ∈ (–b, b)

(D) f(x) . f(–x) = –1 for x ∈ (–b, b)

3) Interval to which k may belong so that function $g(x) = \left(1 - \frac{\sqrt{21 - 4k - k^2}}{k+1}\right) x^3 + 5x + \sqrt{6}$ is increasing at every point of its domain is

(A) (2, 3]

(B) $[-7, -1)$

(C) $[-6, -2]$

(D) (2, 2.5]

SECTION-II

1) Let the angle of intersection of the curves y = [|sinx| + |cosx|] and xy = 2, (where [.] denotes

greatest integer function) is θ then $|\tan\theta|$ is

2) The differential coefficient of $\sin^{-1}\left(\frac{4\sin 2x + 3\cos 2x}{5}\right)$ w.r.t. $\cos^{-1}\left(\frac{5\cos x - 4\sin x}{\sqrt{41}}\right)$ is k then 2k equals

3) Let $f : [0, 2] \to \mathbb{R}$ be the function defined by
$f(x) = (3 - \sin(2\pi x))\sin\left(\pi x - \frac{\pi}{4}\right) - \sin\left(3\pi x + \frac{\pi}{4}\right)$
If $\alpha, \beta \in [0, 2]$ are such that $\{x \in [0, 2] : f(x) \geq 0\} = [\alpha, \beta]$, then the value of $\beta - \alpha$ is

4) If $\alpha = \frac{\pi}{31}$ then the value of $S = \frac{1}{\cos\alpha} + \frac{2\cos\alpha}{\cos 2\alpha} + \frac{4\cos 3\alpha}{\cos 4\alpha} + \frac{8\cos 7\alpha}{\cos 8\alpha} + \frac{16\cos 15\alpha}{\cos 16\alpha}$

5) $\lim_{x\to 0}\left[\ell n\left(1 + \sin^2 x\right) . \cot\left(\ell n^2\ (1 + x)\right)\right]$

6) The number of points, where the function
$f : \mathbb{R} \to \mathbb{R}$, $f(x) = |x-1|\cos|x-2|\sin|x-1| + (x-3)\left|x^2 - 5x + 4\right|$, is NOT differentiable, is

SECTION-I (iii)

1) Let $\sum_{r=1}^{\infty} a_{2r} = 6$ & $\sum_{r=1}^{\infty} a_{2r-1} = 9$, where a_n is n^{th} term of an infinite G.P., then

List-I		List-II	
(P)	Common ratio + first term is equal to	(1)	45
(Q)	a_5 is equal to	(2)	$\frac{17}{3}$
(R)	$\sum_{r=1}^{\infty} a_r^2$ is equal to	(3)	$\frac{80}{81}$
(S)	$\sum_{r=1}^{11} r\frac{a_{2r}}{a_{2r-1}}$ is equal to	(4)	44
		(5)	0

(A) P → 2;Q → 5;R → 1;S → 3

(B) P → 5;Q → 3;R → 2;S → 4

(C) P → 2;Q → 3;R → 1;S → 4

(D) P → 4;Q → 1;R → 5;S → 2

2) If $f(x) = \sin^{-1}\left(\frac{4x}{4+x^2}\right)$, then

List-I		List-II	
(P)	$f\left(\sqrt{8}-2\right)$	(1)	$\pi + \tan^{-1}\left(\frac{4\pi}{4-\pi^2}\right)$
(Q)	$f\left(-\frac{3}{\sqrt{8}-2}\right)$	(2)	$-\pi + 2\tan^{-1}\left(\frac{3}{4}\left(\sqrt{2}+1\right)\right)$
(R)	$f\left(\sqrt{12}-\sqrt{3}\right)$	(3)	$\tan^{-1}\frac{4\pi}{\pi^2-4}$
(S)	f(π)	(4)	$\frac{\pi}{2} - 2\tan^{-1}\left(\sqrt{2}-1\right)$
		(5)	$\frac{\pi}{2} - \tan^{-1}\left(\frac{\sqrt{3}}{12}\right)$

(A) P → 4;Q → 2;R → 5;S → 3

(B) P → 5;Q → 4;R → 2;S → 1

(C) P → 2;Q → 4;R → 5;S → 1

(D) P → 1;Q → 2;R → 4;S → 3

3) Match the following :

List-I		List-II	
(P)	f : $R^+ \to [2, \infty)$ and $f(x) = 2 + 3x^2$ then f(x) is	(1)	one-one
(Q)	f : R → R and $f(x) = \frac{e^{x^2} - e^{-x^2}}{e^{x^2} + e^{-x^2}}$ then f(x) is	(2)	many-one
(R)	f : $(-\infty, -1] \to \left[e^6, \infty\right)$ and $f(x) = e^{x^2-3x+2}$ then f(x) is	(3)	onto
(S)	f : A → A and f(f(x)) = x ∀ x ∈ A then f(x) is	(4)	into
		(5)	even function

(A) P → 3;Q → 4;R → 1;S → 5

(B) P → 4;Q → 5;R → 3;S → 4

(C) P → 1;Q → 4;R → 1;S → 3

(D) P → 2;Q → 3;R → 1;S → 3

ANSWER KEYS

PART 1 : PHYSICS

SECTION-I (i)

Q.	1	2	3	4
A.	B	A	B	A

SECTION-I (ii)

Q.	5	6	7
A.	A,C	A,B,C	A,D

SECTION-II

Q.	8	9	10	11	12	13
A.	5.00	3.00	7.00	8.00	15.00	10.00

SECTION-I (iii)

Q.	14	15	16
A.	B	B	C

PART 2 : CHEMISTRY

SECTION-I (i)

Q.	17	18	19	20
A.	D	A	C	D

SECTION-I (ii)

Q.	21	22	23
A.	A,B,C,D	B,C,D	A,B,C,D

SECTION-II

Q.	24	25	26	27	28	29
A.	4.00	40.00	1.25	9.20	5.00	25

SECTION-I (iii)

Q.	30	31	32
A.	A	B	B

PART 3 : MATHEMATICS

SECTION-I (i)

Q.	33	34	35	36
A.	**A**	**D**	**C**	**C**

SECTION-I (ii)

Q.	37	38	39
A.	**A,B,C**	**A,C**	**A,B,C,D**

SECTION-II

Q.	40	41	42	43	44	45
A.	**0.50**	**4.00**	**1.00**	**0.00**	**1.00**	**2.00**

SECTION-I (iii)

Q.	46	47	48
A.	**C**	**A**	**C**

SOLUTIONS

PART 1 : PHYSICS

1)

$a = -4\ m/s^2$; $u = 6\ m/s$; $t = 2\ sec$; $v = 6 - 4 \times 2 = -2\ m/s$

2)

$v^2 = u^2 + 2as$; $0 = v^2 - 2g\sin\theta \times S$,

$S = \left(\frac{v^2}{2g\sin\theta}\right)$, From triangle $\sin\theta = h/S$,

$h = \frac{v^2}{2g}$

3) $x = \sqrt{2}\cos 45^\circ = 1m$

$y = \left(\sqrt{2}\sin 45^\circ - 0.8\right) = 0.2m$

4) For x direction

$u=0$, $a_x=0$, $\Delta s_x=0$; $\Rightarrow \vec{x}_f = \vec{x}_i$, $x_f=2$

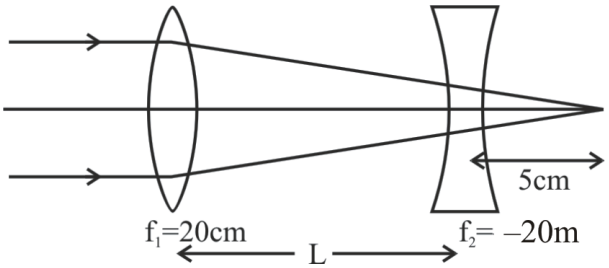
For y direction $u=3$, $a_y=0$, $t=1$

$\Delta s_{yt}=3 \Rightarrow y_f - y_i=3$

For z direction:

$u = 0$, $a_2 = 10$, $s_2 = 5$, $5 = \frac{1}{2}(10)t^2 \Rightarrow t = 1$

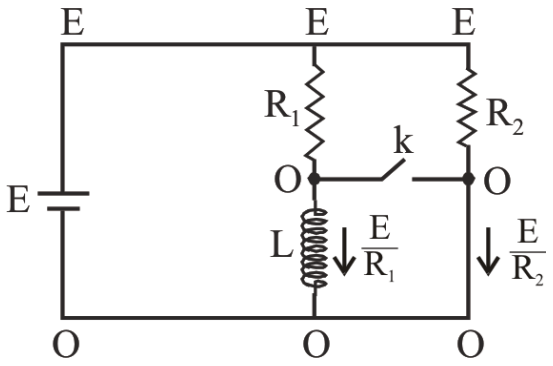
5)



$E_2 : \frac{1}{v_2} - \frac{1}{u_2} = \frac{1}{f} \Rightarrow \frac{1}{5} - \frac{1}{u_2} = \frac{1}{-20}$

$\Rightarrow u_2 = +4$ cm

$\Rightarrow L = 16$ cm

6) 

7) To get zero magnetic field due to ring and solenoid must be opposite to each other and equal to other.

$B_{solenoid} = B_{ring}$

$$\mu_0 n i_{solenoid} = \frac{\mu_0 i_{ring}}{2R}$$

$$4\pi \times 10^{-7} \times \frac{1300}{0.65} \times I = \frac{4\pi \times 10^{-7} \times 8}{2 \times 0.02}$$

$20\ I = 2$

$$I = \frac{1}{10}\text{Amp} = 100\ \text{mA}$$

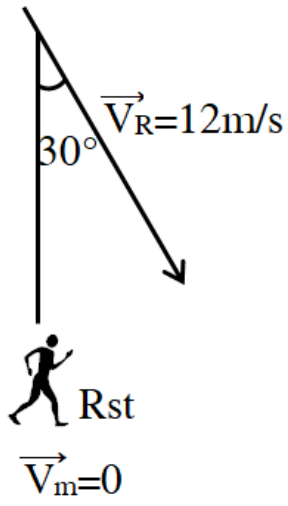
If current is revered then direction changes and $B_{net} = B_1 + B_2$

$$B_{net} = 2B \Rightarrow \frac{\mu_0 i}{R} = \frac{4\pi \times 10^{-7}}{0.02} \times 8$$

$= 16\pi \times 10^{-5}$ Tesla

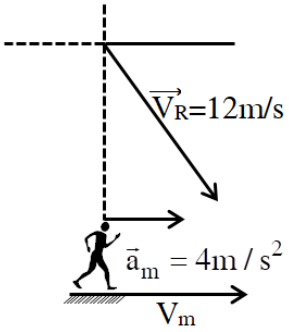
8) $\mu mg = \frac{mv^2}{R}$

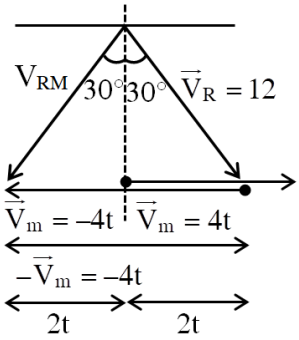
9) Initially **Case-1 :**



Now

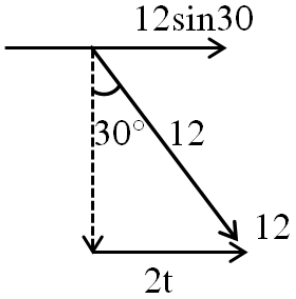
Case-2 :





⇒

From Geometry :



12sin30° = 2t

$12 \times \frac{1}{2} = 2t$

6 = 2t

t = 3 seconds

10)

f = mg sin θ. ... (1)

N = mg cos θ ... (2)

tan θ = μ = θ

$\mu = \frac{3x^2}{6}$ $\frac{x^2}{2} = \frac{1}{2}$

x = 1m y = 1/6 m

11) $\vec{A} \cdot \vec{B} = 0 \Rightarrow 8 + 24 - 4x = 0$

$x = \frac{32}{4} = 8$

12)

$[L] = [ML^2T^{-2}]^a [T]^b [ML^{-3}]^c$

$[L] = [M^{a+c}]\,[L^{2a-3c}]\,[T^{-2a+b}]$

$a + c = 0$

$2a - 3c = 1$

$-2a + b = 0$

$2a + 3a = 1$

$5a = 1$

$a = \frac{1}{5}$

$c = -\frac{1}{5}$

$b = +\frac{2}{5}$

$\Rightarrow \quad x = 5$

$y = +5$

$x = 5$

13) Pi = pf

$m_1 v_1 = (m_1 + m_2)\, v$

$v_1 = 10$ v

$a = \frac{vdv}{dx}$

$\int_0^1 xdx = \int_{v_1}^0 vdv$

$v = 1$

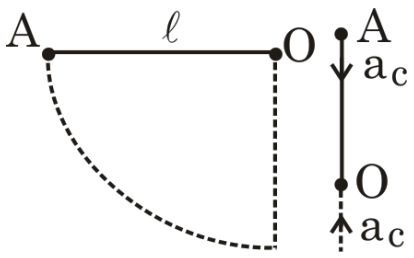
14) (A) Friction along tangent only.

(B) Friction provide centripetal and tangential acceleration.

(C) Friction provide only centripetal acceleration.

(D) If speed is not equal to $\sqrt{rg\tan\theta}$ then friction is along the incline of road to avoid slipping & if speed is equal to $\sqrt{rg\tan\theta}$, no friction.

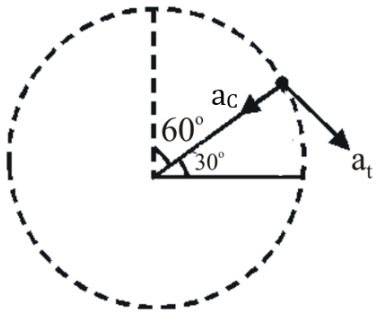
15)



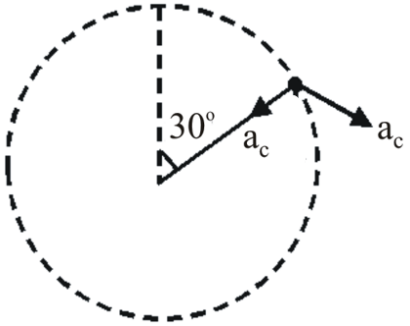
(P) At highest point → Acceleration vertically downward

(Q) At Lowest point → Vertically upward

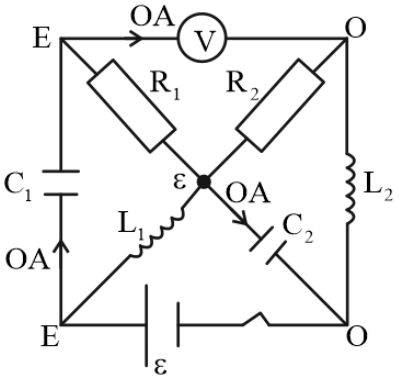
(R) At θ = $\tan^{-1}(\sqrt{3})$ with vertical θ = 60º



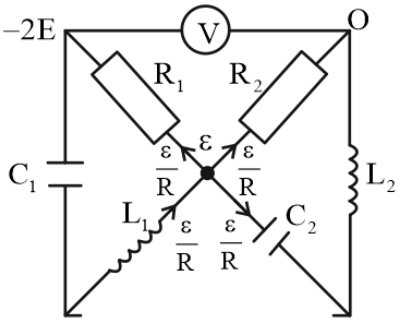
→ Acceleration have both Horizontal and vertical component.



(S) θ = 30º → Acceleration have both.
Horizontal and vertical component.
P → 3, Q → 2, R → 4, S → 4

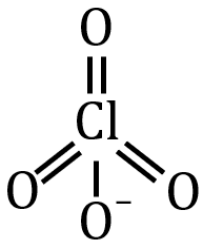
16) 

At steady state



Just after t = 0

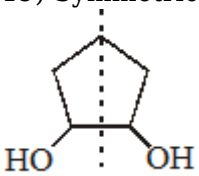
PART 2 : CHEMISTRY

17) 

B.O = $\frac{\text{Number of Bonds}}{\text{Numer of Re sonating St.}} = \frac{7}{4}$

18) With increase in branching boiling point decreases. This is due to decrease in the magnitude of van der Waals forces with increase in branching.
Therefore, boiling point of n-hexane is higher n 2-methyl pentane and 2,2-dimethylbutane.

19) Symmetrical molecule

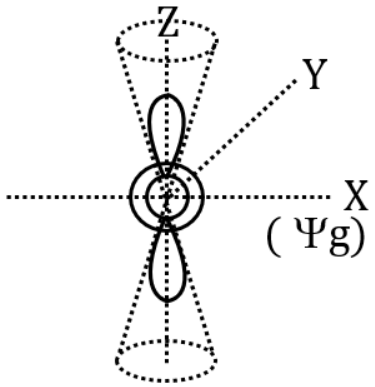


Number of chiral carbon atoms = 2(even)
Number of stereoisomers = $2^{n-1} + 2^{\frac{n}{2}-1}$
$= 2^{n-1} + 2^{\frac{n}{2}-1} = 2^t - 2^0 = 2 + 1 = 3$

20) In (x) both $-CH_3$ are in anti-orientation while in (Y) both $-CH_3$ are in Gauche orientation.
so stable ; x > y (repulsion factor)

21)

It has two angular nodes modified in shape of nodal cones.
One lobe is dumbbell shape while the other is in the shape of circular ring (in XY plane)



22) * no of radial node = n – *l* – 1 = 2 – 0 – 1 = 1
* Probability density ψ^2
* Since ψ contain no ψ or ϕ terms – probability of finding electron do not depend upon direction
* Also since at r = 0, $\psi = \left(\frac{1}{4\sqrt{2}}\right)\left(\frac{1}{a_0}\right)^{3/2}$.(2) ;
ψ^2 is non zero at r = 0

* ψ = 0 when the term $\left(2 - \frac{r}{a_0}\right)$ vanishes putting $2 - \frac{r}{a_0} = 0$
⇒ $r = 2a_0$
at $r = 2a_0$, ψ = 0 : Corresponds to radial node.

23)

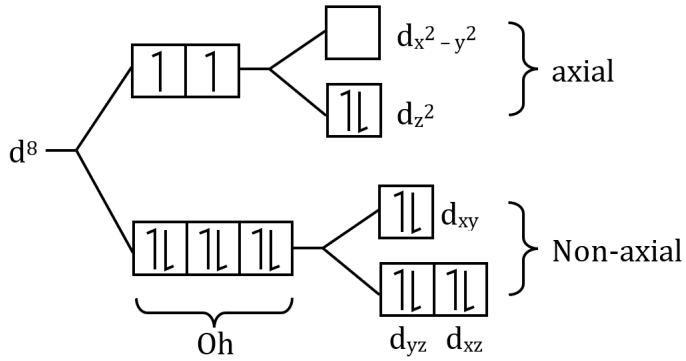
(A) The molecularity of an elementary reaction indicates how many reactant species take part in the step. Thus in unimolecular reaction, only one molecule participates in the step and in bimolecular reaction, two molecules participate in the step.
(B) The rate law of an elementary relation can be predicted by simply seeing the stoichiometry

of reaction. Thus, for the reaction $2A + B \rightarrow C$, the rate law can be written as rate = $k[A]^2[B]$
(C) The slowest elementary step in sequence of the reactions governs the overall rate of formation of product. This step is called rate determining step.
(D) A rate law is often derived from a proposed mechanism by imposing the steady state approximation or assuming that there is a pre-equilibrium.

24) I, IV, VI, VII

25) Average rate in first 10 min. = $\frac{\Delta p}{\Delta t} = \frac{2}{10}\frac{atm}{min}$ Average rate in next 10 min. = $\frac{\Delta p}{\Delta t} = \frac{2}{10}\frac{atm}{min}$
 Reaction is zero order
 $t = \frac{8}{K} = \frac{8}{1/5} = 40\ mins$

26) $Ni^{+2} = d^8$

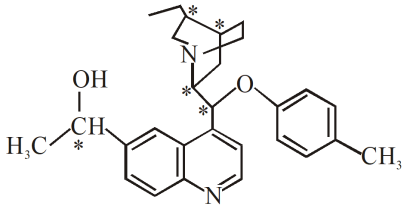


x = 6, y = 2
$\frac{x^2+y^2}{x^2-y^2} = \frac{36+4}{36-4} = \frac{40}{32} = \frac{10}{8} = 1.25$

27) $[Fe(H_2O)_6]_3\ [Fe(CN)_6]_2 \rightarrow 3[Fe(H_2O)_6]^{2+} + 2[Fe(CN)_6]^{3-}$
x = 2.8
y = 4
z = 2.4

28)

No. of chiral centres = 5



29) $ee = \frac{\alpha_{mix}}{\alpha_{pure}} \times 100$

$= \frac{50}{100} \times 100 = 50\%$

% of minor enantiomer $= \frac{50}{2} = 25\%$

30)

(P) $[Cr(NH_3)_6]^{3+}$

$NH_3 \rightarrow SFL$

$Cr^{+3} \rightarrow 3d^3$

d^2sp^3

$-1.2\ \Delta_o$

n = 3

Paramagnetic

(Q) $[Cu(NH_3)_6]^{2+}$

$NH_3 \rightarrow WFL$

$Cu^{+2} \rightarrow 3d^9$

sp^3d^2

$-0.6\ \Delta_o$

n = 1

Paramagnetic

(R) $[Fe(H_2O)_6]^{3+}$

$H_2O \rightarrow WFL$

$Fe^{+3} \rightarrow 3d^5$

sp^3d^2

$-2.0\ \Delta_o$

n = 5

Paramagnetic

(S) $[IrF_6]^{3-}$

$F^- \rightarrow SFL$

$Ir^{+3} \rightarrow 5d^6$

d^2sp^3

$-2.4\ \Delta_o$

n = 0

Diamagnetic

31) (P) $t_{1/2} = \frac{0.586}{k}\sqrt{a} \Rightarrow$ Half order Reaction

$k = \frac{2}{t}\left[\sqrt{a} - \sqrt{a-x}\right]$

(Q) $t_{1/2} = \frac{0.693}{k} \Rightarrow 1^{st}$ order Reaction

$k = \frac{2.303}{t}\log\frac{a}{a-x}$

(R) $t_{1/2} = \frac{a}{2k} \Rightarrow$ zero order Reaction

$$k = \frac{x}{t}$$

(S) $t_{1/2} = \frac{1}{ka} \Rightarrow 2^{nd}$ order Reaction

$$k = \frac{1}{t}\left[\frac{1}{a-x} - \frac{1}{a}\right]$$

32) Pt, H_2(g, 1 atm) | H^+(aq. 0.1M) || Cl^-(aq. 0.01 M) | Cl_2(g, 1 atm)

Cell reaction : $H_2(g) + 2Cl^-(aq) \rightarrow 2H^+(aq) + Cl_2(g)$

$$E = E^o - \frac{0.06}{2}\log\frac{[H^+]^2 P_{Cl_2}}{[Cl^-]^2 P_{H_2}}$$

$$E^o - \frac{0.06}{2}\log\frac{\left(10^{-5}\right)^2}{\left(10^{-2}\right)^2} - E^o + \frac{0.06}{2}\log\frac{\left(10^{-1}\right)^2}{\left(10^{-2}\right)^2}$$

(P) $\Delta E = E_2 - E_1 =$

$$= \frac{8}{2} \times 0.06 = 0.24$$

similarly we have to solve the other equations

(Q) $\Delta E = E_2 - E_1 = -\frac{0.06}{2}\log\frac{(0.1)^2}{(0.01)^2} = -0.06$ V

(R) $\Delta E = E_2 - E_1 = -\frac{0.06}{2}\log\frac{1}{10} = 0.03$ V

(S) $\Delta E = E_2 - E_1 = -\frac{0.06}{2}\log\frac{1}{0.1} = -0.03$ V

PART 3 : MATHEMATICS

33) Number of ways $= \frac{7!}{2!2!}\left[{}^{10}C_7 - \left\{\left({}^9C_7 + {}^8C_7 + {}^7C_7\right) - \left({}^7C_7\right)\right\}\right]$

$= 1260[120 - 44]$

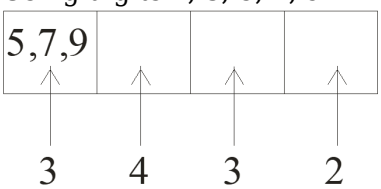
$= 1260 \times 76$

$= 95760$

34)

Numbers between 5000 & 10000

Using digits 1, 3, 5, 7, 9



Total Numbers $= 3 \times 4 \times 3 \times 2 = 72$

35) Applying $C_1 \rightarrow C_1 - C_2$ and then $R_1 \rightarrow R_1 - R_2$ and $R_2 \rightarrow R_2 - R_3$ we will get

$\Rightarrow a - 2b + c = 0$
$\Rightarrow P \equiv (1,-2)$

36) $A^T = A$
$d = |2A^T + AA^T + adjA|$
$2A^T + AA^T + adjA = 2A + A^2 + adjA$
$= \begin{bmatrix} 2\sqrt{3} & 2 & 0 \\ 2 & -2\sqrt{3} & 0 \\ 0 & 0 & 4 \end{bmatrix} + \begin{bmatrix} 4 & 0 & 0 \\ 0 & 4 & 0 \\ 0 & 0 & 4 \end{bmatrix} + \begin{bmatrix} -2\sqrt{3} & -2 & 0 \\ -2 & 2\sqrt{3} & 0 \\ 0 & 0 & -4 \end{bmatrix}$
$= 4I \qquad d = |4I| = 64$
$\sqrt{d} = 8$

37) $f(x) = \cos^2 x \; e^{\tan x}$
$f'(x) = e^{\tan x}(1 - \sin 2x)$
$f''(x) = e^{\tan x}(-\cos 2x)(2) + (1 - \sin 2x)\, e^{\tan x} \sec^2 x$

$$= e^{\tan x}\left(\frac{2\left(\tan^2 x - 1\right)}{1 + \tan^2 x} + \sec^2 x - 2\tan x\right)$$

$$= e^{\tan x}\left(\frac{2\left(\tan^2 x - 1\right)}{1 + \tan^2 x} + (1 - \tan x)^2\right)$$

$= f(x)(\tan x - 1)(\tan^3 x - \tan^2 x + 3\tan x + 1)$

Let $g(x) = \tan^3 x - \tan^2 x + 3\tan x + 1$
$g'(x) = (3\tan^2 x - 2\tan x + 3)\sec^2 x > 0$

so g(x) is increasing in $\left(-\frac{\pi}{4}, 0\right)$.

Also $g\left(-\frac{\pi}{4}\right) g(0) < 0$

g(x) has exactly one root, in $\left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$

 f ″(x) has exactly 2 roots in $\left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$

At $x = \frac{\pi}{4}$ f ′(x) has local minima

38) $f'(x) = \dfrac{f(x)}{\sqrt{b^2 - x^2}} \Rightarrow \ln f(x) = \sin^{-1}\left(\dfrac{x}{b}\right) + c$

$f(x) = e^{\sin^{-1}(x/b)}$

$f'(x) = \dfrac{e^{\sin^{-1}(x/b)}}{\sqrt{1 - (x/b)^2}} \cdot \dfrac{1}{b}$

39) Firstly, for g(x) to be defined.
$21 - 4k - k^2 \geqslant 0; \; k^2 + 4k - 21 \leqslant 0$

$(k+7)(k-3)0;\ k \in [-7, 3]$

Now $g' = 3\left(1 - \frac{\sqrt{21-4k-k^2}}{k+1}\right)x^2 + 5$

as g(x) is defined for $x \in R$.

domain of g(x) = R

Now g(x) is to be increasing at every point of its domain: $g(x) > 0$ for all $x \in R$.

For this to be true, coefficient of $x^2 > 0$

$$1 - \frac{\sqrt{21-4k-k^2}}{k+1} > 0$$

$$\frac{\sqrt{21-4k-k^2}}{k+1} < 1$$

Case 1 : If $k+1 > 0 :\ k > -1$,

Then $\left(\sqrt{21-4k-k^2}\right)^2 < (k+1)^2$

$21-4k-k^2 < k^2+1+2k$

$2k^2+6k-20 > 0$

$k^2+3k-10 > 0$

$\Rightarrow (k+5)(k-2) > 0$

$+ \quad - \quad +$

$-5 \quad 2$

$k \in (-\infty,\ -5) \cup (2,\ \infty)$

Combining all we get : $k \in (2, 3]$

Case 2 : If $k+1 < 0,\ k < -1$

Then LHS will become –ve

Hence inequality holds true for all $k < -1$

Combining all we get $k \in [-7, -1)$

Taking union: $k \in [-7, -1) \cup (2, 3]$

A, B, C, D are correct

40) $|\sin x| + |\cos x| = \sqrt{\sin^2 x + \cos^2 x + 2|\sin x||\cos x|}$

$= \sqrt{1 + |\sin 2x|} \geq 1$

 y = [|sinx| + |cosx|] = 1

Now, point of intersection of y = 1 & xy = 2 is (2, 1)

$y = \frac{2}{x}$

$\frac{dy}{dx} = -\frac{2}{x^2} \quad \therefore \quad \left.\frac{dy}{dx}\right|_{x=2} = -\frac{1}{2}$

$m_2 = 0$ {slope of y = 1}

 Angle of intersection is $\theta = \tan^{-1}\frac{1}{2}$

41) Let $u = \sin^{-1}\left(\frac{4}{5}\sin 2x + \frac{3}{5}\cos 2x\right)$

$u = \sin^{-1}\sin(\alpha + 2x)$

$\Rightarrow u = \alpha + 2x$

where $\sin\alpha = \frac{3}{5}, \cos\alpha = \frac{4}{5}$

Similarly, $v = \cos^{-1}\left(\frac{5\cos x - 4\sin x}{\sqrt{41}}\right)$

$= \cos^{-1}\cos(\beta + x)$,

Here, $\cos\beta = \frac{5}{\sqrt{41}}, \sin\beta = \frac{4}{\sqrt{41}}$

i.e.

$\nu = \beta + x$

$\therefore \frac{du}{dv} = \frac{\frac{du}{dx}}{\frac{dv}{dx}} = \frac{2}{1} = 2 = k$

$2k = 4$

42) Let $\pi x - \frac{\pi}{4} = \theta \in \left[\frac{-\pi}{4}, \frac{7\pi}{4}\right]$

So, $\left(3 - \sin\left(\frac{\pi}{2} + 2\theta\right)\right)\sin\theta \geqslant \sin(\pi + 3\theta)$

$\Rightarrow (3 - \cos2\theta)\sin\theta \geq -\sin3\theta$

$\Rightarrow \sin\theta[3 - 4\sin^2\theta + 3 - \cos2\theta] \geq 0$

$\Rightarrow \sin\theta(6 - 2(1 - \cos2\theta) - \cos2\theta) \geq 0$

$\Rightarrow \sin\theta(4 + \cos2\theta) \geq 0$

$\Rightarrow \sin\theta \geq 0$

$\Rightarrow \theta \in [0, \pi] \Rightarrow 0 \leqslant \pi x - \frac{\pi}{4} \leqslant \pi$

$\Rightarrow x \in \left[\frac{1}{4}, \frac{5}{4}\right]$

$\Rightarrow \beta - \alpha = 1.$

43) $S = \frac{2\sin\alpha}{\sin2\alpha} + \frac{2\cos\alpha}{\cos2\alpha} + \ldots\ldots\ldots$

$S = \frac{4\sin3\alpha}{\sin4\alpha} + \frac{4\cos3\alpha}{\cos4\alpha} + \frac{8\cos7\alpha}{\cos8\alpha} + \frac{16\cos15\alpha}{\cos16\alpha}$

$S = \frac{8\sin7\alpha}{\sin8\alpha} + \frac{8\cos7\alpha}{\cos8\alpha} + \frac{16\cos15\alpha}{\cos16\alpha}$

$= \frac{16\sin15\alpha}{\sin16\alpha} + \frac{16\cos15\alpha}{\cos16\alpha}$

$\frac{32\sin31\alpha}{\sin32\alpha} = \frac{32\sin\pi}{\sin32\pi/31} = 0$

44) $\lim_{x\to0} \ell n\left(1 + \sin^2x\right).\cot\left(\ell n^2(1 + x)\right)$

$\Rightarrow \lim_{x\to0} \frac{\ell n\left(1 + \sin^2x\right)}{\sin^2x} \times \sin^2x \times \frac{\ell n^2(1 + x)}{\tan\left(\ell n^2(1 + x)\right) \times \ell n^2(1 + x)}$

$\{\sin^2x \to 0 \text{ and } \ell n(1 + x) \to 0 \text{ as } x \to 0\}$

$\Rightarrow \lim_{x\to0} \frac{\sin^2x}{\ell n^2(1 + x)}$

$\Rightarrow \lim_{x\to0} \frac{\sin^2x}{x^2} \times \left(\frac{x}{\ell n(1 + x)}\right)^2$

$\left\{\lim_{x\to0} \frac{\sin x}{x} = 1 \text{ and } \lim_{x\to0} \frac{\ell n(1 + x)}{x} = 1\right\}$

$\Rightarrow 1 \times 1 = 1$

45) $f : R \to R$

$f(x) = |x-1|\cos|x-2|\sin|x-1| + (x-3)\left|x^2-5x+4\right|$

$= |x-1|\cos|x-2|\sin|x-1| + (x-3)|x-1||x-4|$

$= |x-1|[\cos|x-2|\sin|x-1| + (x-3)|x-4|]$

Sharp edges at $x = 1$ and $x = 4$

$\therefore$ Non-differentiable at $x = 1$ and $x = 4$

46) Let first term & common ratio of G.P. are A & R

Given that $AR + AR^3 + AR^5 + \ldots \infty = 6$

& $A + AR^2 + AR^4 + \ldots \infty = 9$

$\Rightarrow R(A + AR^2 + AR^4 + \ldots \infty) = 6$

$\Rightarrow R = \frac{6}{9} = \frac{2}{3}$ & $\frac{A}{1-R^2} = 9 \Rightarrow A = 5$

(P) $A + R = \frac{2}{3} + 5 = \frac{17}{3}$

(Q) $a_5 = A.R^4 = 5.\frac{16}{81} = \frac{80}{81}$

(R) $\sum_{r=1}^{\infty} a_r^2 = \frac{A^2}{1-R^2} = \frac{25}{1-\frac{4}{9}} = \frac{25 \times 9}{5} = 45$

(S) $\sum_{r=1}^{11} r\frac{a_{2r}}{a_{2r-1}} = \sum_{r=1}^{11} rR = \frac{2}{3}\sum_{r=1}^{11} r = \frac{2}{3}.\frac{11 \times 12}{2} = 44$

47) $$\sin^{-1}\left(\frac{4x}{4+x^2}\right) = \begin{cases} -\pi - 2\tan^{-1}\left(\frac{x}{2}\right) & ; \quad x < -2 \\ 2\tan^{-1}\left(\frac{x}{2}\right) & ; \quad -2 \leqslant x \leqslant 2 \\ \pi - 2\tan^{-1}\left(\frac{x}{2}\right) & ; \quad x > 2 \end{cases}$$

48) (P) $f(x) = 3x^2 + 2 \in (2,\infty)$

for $x \in R^+$ $3x^2 + 2$ is one-one and into

$\Rightarrow$ (P) $\rightarrow$ (1), (4)

(Q) $f(x) = \frac{e^{x^2} - e^{-x^2}}{e^{x^2} + e^{-x^2}} = \frac{e^{2x^2} - 1}{e^{2x^2} + 1}$

$y = \frac{t^2-1}{t^2+1} \Rightarrow t^2(y-1) + (y+1) = 0$

$t^2 = \frac{-(y+1)}{(y-1)}$

$\frac{-(y+1)}{y-1} \geqslant 0 \Rightarrow \frac{y+1}{y-1} \leqslant 0$

$y \in [-1,1)$

codomain $\neq$ Range $\Rightarrow$ into function many one

$\Rightarrow$ (Q) $\rightarrow$ (4), (2), (5)

(R) $f(x) = e^{x^2-3x+2}$

$x^2 - 3x + 2 \in [6, \infty)$

$e^{x^2-3x+2} \in \left[e^6, \infty\right)$ Hence onto

⇒ R → (1), (3)

(S) f(f(x)) = x

f(x) = $f^{-1}(x)$

y = x

⇒ S → (1), (3)